EXPERIMENT 1

Title: Regular Expressions in Python for Natural Language Processing

Aim:

To write a Python program using regular expressions to identify and extract specific patterns such as phone numbers, email addresses, and numerical values from a given text. Regular expressions are useful in Natural Language Processing (NLP) for text searching, validation, and information extraction.

Algorithm:

1. Start the program.
2. Import the re module, which provides functions for working with regular expressions.
3. Define a sample text containing a phone number, an email address, and numerical values.
4. Use the re.search() function to identify a valid phone number consisting of ten digits.
5. Use another regular expression with re.search() to identify an email address in the text.
6. Apply re.findall() to extract all numerical sequences present in the text.
7. Check whether the phone number and email address are found.
8. Display the extracted phone number, email address, and list of numbers.
9. Stop the program.

Program:

*Exp-01-Regular Expression.py - C:\Users\jerin\OneDrive\Desktop\NLP\Experiments\Exp-01-...

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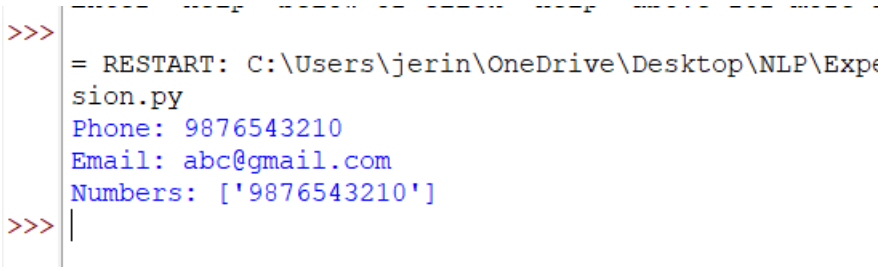
```
import re

text = "Call 9876543210 or email abc@gmail.com"

phone = re.search(r"\b\d{10}\b", text)
email = re.search(r"\S+@\S+\.\S+", text)
numbers = re.findall(r"\d+", text)

print("Phone:", phone.group() if phone else "Not found")
print("Email:", email.group() if email else "Not found")
print("Numbers:", numbers)
```

Output:



Result:

Thus, the Python program to identify and extract phone numbers, email addresses, and numerical values using regular expressions was implemented successfully. The program demonstrates how regular expressions can be used for pattern matching and information extraction in NLP applications.